



SkillBridge

Bridging the Gap Between Skills and Career Opportunities



Project Overview

SkillBridge is an AI-powered career development platform that helps users identify the gap between their current skills and the skills required for their dream job. By leveraging **Google Gemini AI**, **YouTube Data API**, and a structured **Three-Tier Architecture**, the platform provides users with personalized career guidance, learning resources, project recommendations, and progress tracking.

Unlike traditional resume analyzers that provide one-time suggestions, SkillBridge acts as a **long-term career companion**, storing AI-generated insights, learning roadmaps, and user progress so that users can continue their learning journey anytime.



Vision

"Bridge the gap between where you are today and where you aspire to be tomorrow."

SkillBridge aims to simplify career planning by transforming a user's resume into a personalized learning roadmap backed by Artificial Intelligence.



Problem Statement

Many job seekers struggle with questions such as:

- What skills am I missing for my target role?
- Which technologies should I learn first?
- Which courses or videos are actually worth watching?

- What projects should I build for my portfolio?
- How do I track my progress effectively?

Existing platforms usually solve only one of these problems. SkillBridge combines them into one intelligent platform.



Proposed Solution

SkillBridge enables users to:

- Upload their resume.
- Select the job role they aspire to.
- Receive AI-generated skill gap analysis.
- View personalized learning recommendations.
- Follow an AI-generated roadmap.
- Build suggested projects.
- Track learning progress.
- Resume their learning journey anytime.

All generated recommendations are securely stored, allowing users to continue exactly where they left off.



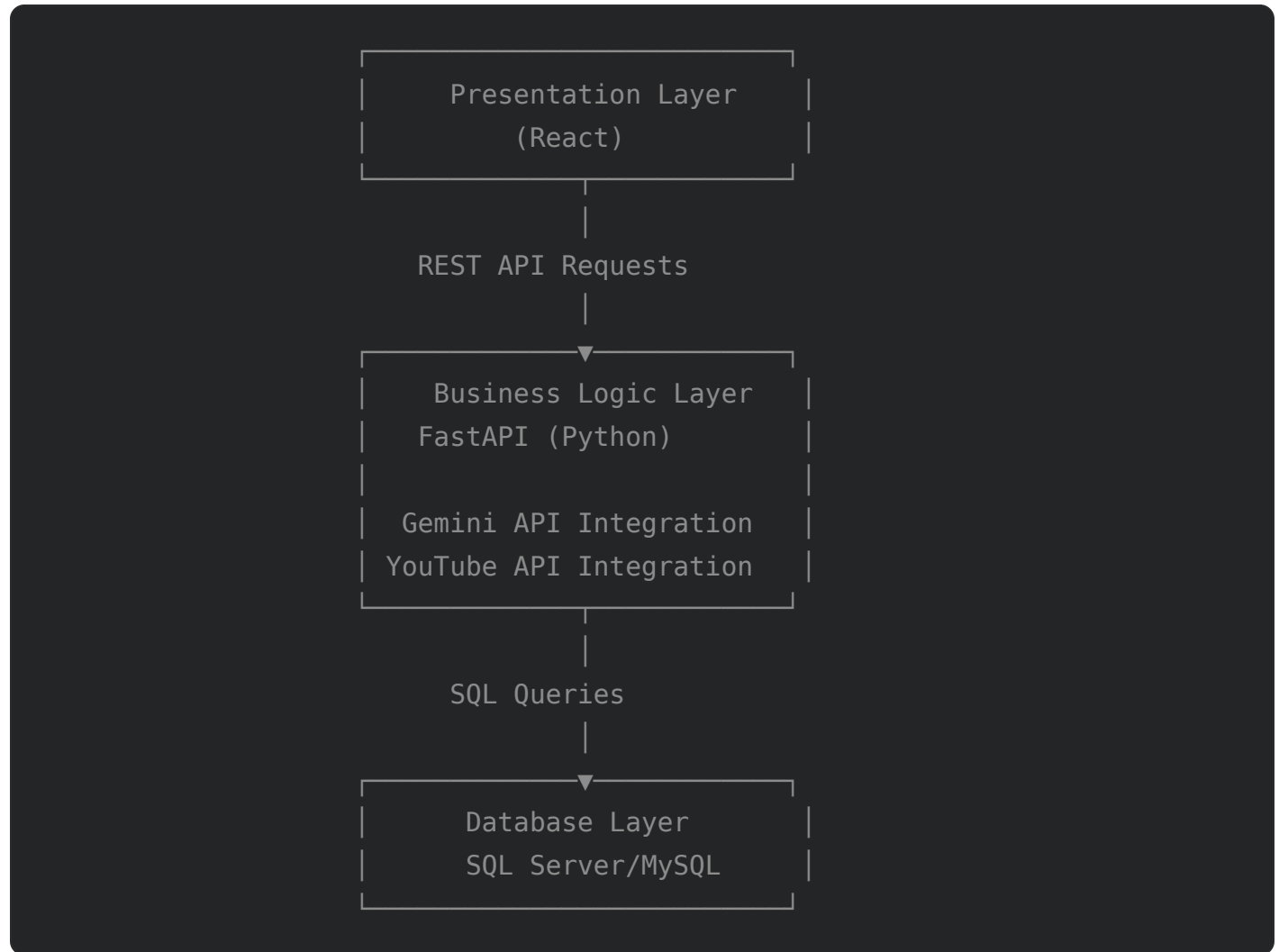
Objectives

The primary objectives of SkillBridge are:

- Analyze resumes using AI.
 - Identify missing skills for a desired career role.
 - Recommend learning resources.
 - Suggest projects based on the user's experience.
 - Generate personalized career roadmaps.
 - Track learning progress.
 - Maintain historical records of user analyses.
 - Minimize repeated AI calls by storing previous responses.
-

System Architecture

SkillBridge follows a **Three-Tier Architecture**.



Technology Stack

Layer	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	FastAPI (Python)
Authentication	JWT
Database	SQL Server

Layer	Technology
AI Engine	Google Gemini API
Video Recommendations	YouTube Data API
Resume Parsing	Python Libraries (PyPDF2 / pdfplumber / python-docx)
Deployment	Docker



User Workflow



↓
▼
Personal Dashboard
↓
▼
Track Progress

🌟 Core Features



User Management

- Secure Registration
- Login Authentication
- JWT-based Sessions
- User Profile



Resume Management

- Upload Resume (PDF/DOCX)
- Multiple Resume Support
- Resume History
- Delete Resume
- Multiple Career Goals



AI Resume Analysis

Gemini analyzes:

- Existing Skills
- Missing Skills
- Resume Strength
- Resume Weaknesses
- Suggested Improvements



Learning Recommendations

The platform recommends:

- Online Courses
 - Official Documentation
 - Learning Resources
 - Technology Guides
-



YouTube Integration

The YouTube API provides:

- Beginner Tutorials
 - Advanced Tutorials
 - Playlists
 - Technology-Specific Videos
-



Personalized Roadmaps

Gemini generates customized learning paths.

Example:

Python



SQL



Git

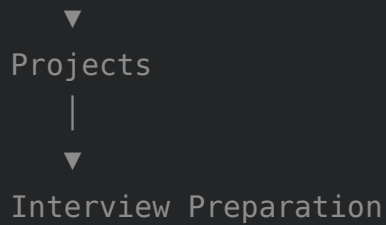


Docker



FastAPI





▼
Projects
|
▼
Interview Preparation

AI Project Recommendations

Projects are categorized into:

- Beginner
- Intermediate
- Advanced

Each project contains:

- Description
- Required Skills
- Technologies
- Estimated Difficulty

Progress Tracking

Users can:

- Mark roadmap steps as complete
- Track project completion
- Monitor learning progress
- View completion percentage

Dashboard

The dashboard displays:

- Active Career Goals
- Resume Status
- Missing Skills

- Learning Roadmap
 - Recommended Videos
 - Suggested Projects
 - Progress Overview
-



Database Design

The system stores:

Users



Resumes



Career Goals



Skill Analysis



Roadmaps



Roadmap Steps



Courses



YouTube Videos

↓

Projects

↓

Project Tasks

↓

Progress

↓

AI Responses

↓

Activity Logs



AI Processing Flow

Resume Upload

↓

Resume Parsing

↓

Gemini Prompt

↓

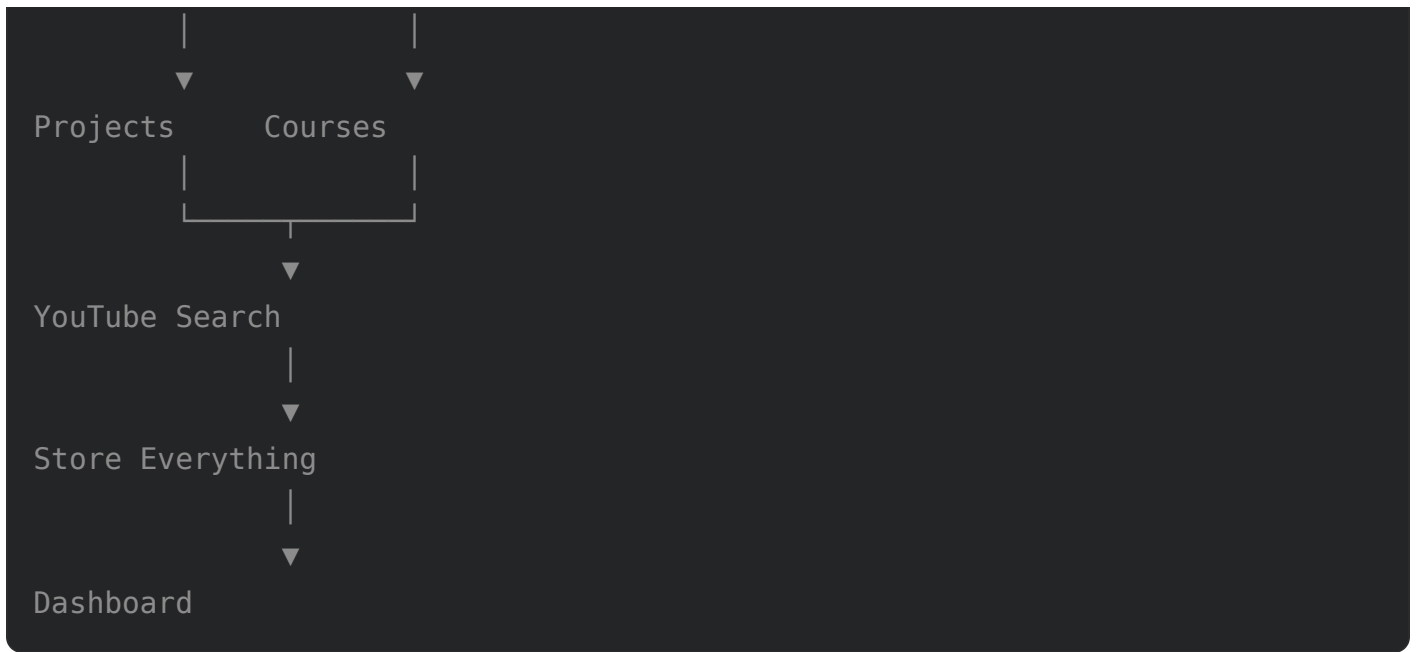
AI Response

↓

↓

Skill Gap

Roadmap



Advantages

- AI-powered career guidance
- Personalized learning experience
- Persistent storage of AI recommendations
- Multiple career goals per user
- Interactive progress tracking
- Reduced API costs through cached AI responses
- Scalable architecture
- Enterprise-ready design



Future Enhancements

Future versions of SkillBridge may include:

- AI Interview Preparation
- Resume Version Comparison
- GitHub Portfolio Analysis
- LinkedIn Profile Integration
- AI Chat Assistant
- Mock Coding Interviews
- Email Notifications

- Calendar Integration
 - Job Portal Integration
 - AI-Based Resume Scoring
 - Vector Database for Context-Aware AI Chat
-

Why SkillBridge?

Unlike conventional resume analyzers, SkillBridge offers an end-to-end career development ecosystem. It not only identifies skill gaps but also provides actionable learning paths, project recommendations, curated learning resources, and continuous progress tracking.

The platform evolves with the user, making career development structured, measurable, and personalized.

Expected Outcome

At the end of the process, every user will have:

-  A detailed AI-powered resume analysis
 -  A list of missing skills
 -  Recommended courses and YouTube videos
 -  A personalized learning roadmap
 -  AI-generated project ideas
 -  A progress tracker
 -  Persistent learning history
 -  Multiple career goals managed from one dashboard
-

SkillBridge

"Your Personalized AI Career Companion — Learn Smarter, Build Better, Get Hired."